Practice Questions for Reduced Costs

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1 Questions

1. In the context of reduced costs, explain the economic interpretation of a reduced cost of zero for a non-basic variable in a linear programming problem.
   a. The non-basic variable's contribution to the objective function is negligible and can be ignored.
   b. A small increase in the non-basic variable will not improve the objective function value.
   c. The current solution is optimal, and there is no incentive to increase the non-basic variable.
   d. The non-basic variable is already at its maximum allowable value given the constraints.
   e. The non-basic variable's reduced cost is undefined in this case.

2. How are reduced costs related to the shadow prices (dual variables) in linear programming?
   a. Reduced costs are the negative of the shadow prices.
   b. Reduced costs and shadow prices are independent and unrelated concepts.
   c. Reduced costs represent the change in the objective function value per unit increase in a basic variable, while shadow prices represent the change in the objective function value per unit increase in a non-basic variable.
   d. Reduced costs are the shadow prices for non-basic variables.
   e. Reduced costs and shadow prices are always equal in magnitude but opposite in sign.

3. Suppose a non-basic variable in a maximization linear programming problem has a negative reduced cost. What does this imply about the potential for improving the objective function value?
   a. The objective function value cannot be improved by changing this variable.
   b. Increasing the value of this variable will improve the objective function value.
   c. Decreasing the value of this variable will improve the objective function value.
   d. The reduced cost is irrelevant to the objective function value.
   e. The problem is infeasible.

4. Explain how reduced costs are used in the simplex method to determine whether a current solution is optimal.
   a. If all reduced costs are positive, the solution is optimal.

b. If any reduced cost is negative, the solution is not optimal.

c. If all reduced costs are zero, the solution is optimal, but there may be alternative optimal solutions.

d. If all reduced costs are non-positive, the solution is optimal for a maximization problem.

e. Reduced costs are not directly used to determine optimality in the simplex method.

5. Explain the concept of reduced costs in linear programming and their significance in interpreting the optimal solution.

   a. Reduced costs represent the shadow prices of the non-basic variables, indicating the marginal increase in the objective function value if a non-basic variable were to enter the basis.

   b. Reduced costs represent the amount by which the objective function coefficient of a non-basic variable must be improved to make that variable enter the optimal basis.

   c. Reduced costs are the values of the slack variables in the optimal solution, indicating the unused resources.

   d. Reduced costs are the differences between the objective function coefficients of basic and non-basic variables, indicating the relative profitability of each variable.

   e. Reduced costs are the values of the dual variables in the optimal solution, indicating the marginal value of each constraint.

6. In a maximization linear programming problem, what is the economic interpretation of a reduced cost of zero for a non-basic variable?

   a. The variable's contribution to the objective function is positive, and increasing it would improve the solution.

   b. The variable's contribution to the objective function is negative, and increasing it would worsen the solution.

   c. The variable's current value is optimal, and any change would not affect the objective function value.

   d. The variable's contribution to the objective function is zero, and increasing it would not affect the objective function value.

   e. The variable's contribution to the objective function is indeterminate, and further analysis is needed.

7. How are reduced costs related to the shadow prices (dual variables) in linear programming?

   a. Reduced costs are the negative of the shadow prices.

   b. Reduced costs are equal to the shadow prices.

   c. Reduced costs and shadow prices are unrelated concepts.

   d. Reduced costs are the absolute values of the shadow prices.

   e. Reduced costs are a linear transformation of the shadow prices, the specific transformation depending on the problem's structure.

8. Suppose a non-basic variable in a maximization linear programming problem has a negative reduced cost. What does this imply about the potential for improving the objective function value?

a. The objective function value cannot be improved.

b. The objective function value can be improved by increasing the non-basic variable.

c. The objective function value can be improved by decreasing the non-basic variable.

d. The objective function value can only be improved by changing other variables.

e. The impact on the objective function value is indeterminate.

9. Explain how reduced costs are used in the simplex method to determine whether a current solution is optimal.

   a. If all reduced costs are positive, the solution is optimal.

   b. If all reduced costs are negative, the solution is optimal.

   c. If any reduced cost is negative, the solution is not optimal.

   d. If any reduced cost is positive, the solution is not optimal.

   e. Reduced costs are not used to determine optimality in the simplex method.

10. In a maximization linear programming problem, a non-basic variable has a reduced cost of -5. What does this indicate?

   a. The current solution is optimal.

   b. Increasing this variable will improve the objective function value.

   c. The variable's coefficient in the objective function is too high.

   d. The variable should be removed from the problem.

   e. The current solution is infeasible.

2 Answers

1. In the context of reduced costs, explain the economic interpretation of a reduced cost of zero for a non-basic variable in a linear programming problem.

   a. While a very small contribution might be implied, a reduced cost of zero doesn't mean the variable is negligible; it means its marginal contribution is zero at the current optimal point.

   b. This is partially true, but it doesn't fully capture the economic interpretation. The key is that the current solution is optimal, and there's no incentive to change it.

   c. A reduced cost of zero for a non-basic variable indicates that the current solution is optimal, and increasing the non-basic variable would not improve the objective function value. The variable is at a point where any change would not improve the solution.

   d. This is incorrect. A non-basic variable with a reduced cost of zero is at a value of zero. It's not necessarily at its maximum allowable value.

   e. The reduced cost is always defined for a non-basic variable; it's a key component of the simplex method's optimality conditions.

2. How are reduced costs related to the shadow prices (dual variables) in linear programming?

   a. Reduced costs are directly related to shadow prices. In fact, the reduced cost of a non-basic variable is the negative of the corresponding dual variable (shadow price) in the optimal dual solution.

   b. Reduced costs and shadow prices are fundamentally linked through the duality theorem and complementary slackness.

   c. This statement reverses the roles of basic and non-basic variables. Shadow prices relate to changes in the RHS of constraints, not directly to changes in non-basic variables.

   d. Reduced costs are related to shadow prices, but they are not the same thing. They are related through the negative sign.

   e. They are related through a negative sign, but not always equal in magnitude. The magnitude depends on the specific problem and solution.

3. Suppose a non-basic variable in a maximization linear programming problem has a negative reduced cost. What does this imply about the potential for improving the objective function value?

   a. This is only true if the reduced cost is zero or positive.

   b. A negative reduced cost for a non-basic variable in a maximization problem indicates that increasing this variable will improve the objective function value. The magnitude of the reduced cost represents the rate of improvement.

   c. Decreasing a non-basic variable (which is already at zero) would not change the objective function value.

   d. The reduced cost is crucial for determining whether the current solution is optimal and how to improve it if it is not.

   e. A negative reduced cost does not imply infeasibility.

4. Explain how reduced costs are used in the simplex method to determine whether a current solution is optimal.

a. This is incorrect for a maximization problem. A negative reduced cost indicates a non-optimal solution.

b. This is partially correct, but it doesn't cover the case where all reduced costs are zero.

c. This is partially correct, but it only addresses the case of zero reduced costs. Negative reduced costs also need to be considered.

d. In a maximization problem, if all reduced costs are non-positive (zero or negative), the current solution is optimal. A negative reduced cost indicates a potential for improvement.

e. Reduced costs are a fundamental part of the simplex method's optimality conditions.

5. Explain the concept of reduced costs in linear programming and their significance in interpreting the optimal solution.

   a. Shadow prices are represented by the dual variables, not the reduced costs. While related, they provide different information.

   b. Reduced costs indicate how much the objective function coefficient of a non-basic variable needs to improve (increase in maximization, decrease in minimization) before that variable becomes part of the optimal solution. A negative reduced cost for a non-basic variable in a maximization problem, for example, means that increasing its objective function coefficient would improve the objective function value.

   c. Slack variables indicate unused resources; reduced costs relate to the potential for improvement by including non-basic variables.

   d. Reduced costs are not simply differences between objective function coefficients; they account for the impact of the variable on the entire system of constraints.

   e. Dual variables represent shadow prices, which are the marginal values of the constraints. Reduced costs are associated with variables, not constraints.

6. In a maximization linear programming problem, what is the economic interpretation of a reduced cost of zero for a non-basic variable?

   a. A positive contribution would imply a positive reduced cost, indicating that increasing the variable would improve the solution.

   b. A negative contribution would imply a negative reduced cost, indicating that increasing the variable would worsen the solution.

   c. While the current value is optimal, the statement is too strong. A reduced cost of zero indicates that a small change would not affect the objective function value, but larger changes might.

   d. A reduced cost of zero for a non-basic variable in a maximization problem means that increasing the variable from its current value of zero would not change the objective function value. The variable is already at its optimal level.

   e. The contribution is not indeterminate; it's precisely zero.

7. How are reduced costs related to the shadow prices (dual variables) in linear programming?

   a. Reduced costs for non-basic variables in the primal problem are the negative of the shadow prices (dual variables) in the corresponding dual problem.

   b. They are not equal; there's a negative sign relationship.

c. They are closely related through duality theory.

d. The relationship is not simply taking the absolute value.

e. While a linear transformation exists, it's simply negation.

8. Suppose a non-basic variable in a maximization linear programming problem has a negative reduced cost. What does this imply about the potential for improving the objective function value?

   a. This is only true if all reduced costs are non-negative.

   b. A negative reduced cost for a non-basic variable in a maximization problem indicates that increasing that variable would improve the objective function value.

   c. Decreasing a non-basic variable (already at zero) would not change its value.

   d. While changing other variables might improve the solution, this specific non-basic variable also has the potential to improve it.

   e. The impact is clearly defined by the sign of the reduced cost.

9. Explain how reduced costs are used in the simplex method to determine whether a current solution is optimal.

   a. Positive reduced costs for non-basic variables indicate that the current solution is optimal.

   b. Negative reduced costs indicate that the solution is not optimal.

   c. In a maximization problem, if any reduced cost is negative, it indicates that the objective function can be improved by bringing the corresponding non-basic variable into the basis. Therefore, the current solution is not optimal.

   d. Positive reduced costs indicate optimality for non-basic variables.

   e. Reduced costs are a fundamental part of the simplex method's optimality check.

10. In a maximization linear programming problem, a non-basic variable has a reduced cost of -5. What does this indicate?

    a. A negative reduced cost indicates that the current solution is not optimal. There is room for improvement.

    b. A negative reduced cost for a non-basic variable in a maximization problem means that increasing that variable will increase the objective function value. The current solution is not optimal.

    c. The reduced cost is related to the variable's contribution to the objective function considering the constraints, not just its coefficient.

    d. Removing a variable is not the appropriate action; instead, the simplex method would bring this variable into the basis.

    e. Reduced costs are used to assess optimality, not feasibility. Feasibility is determined by whether the constraints are satisfied.